



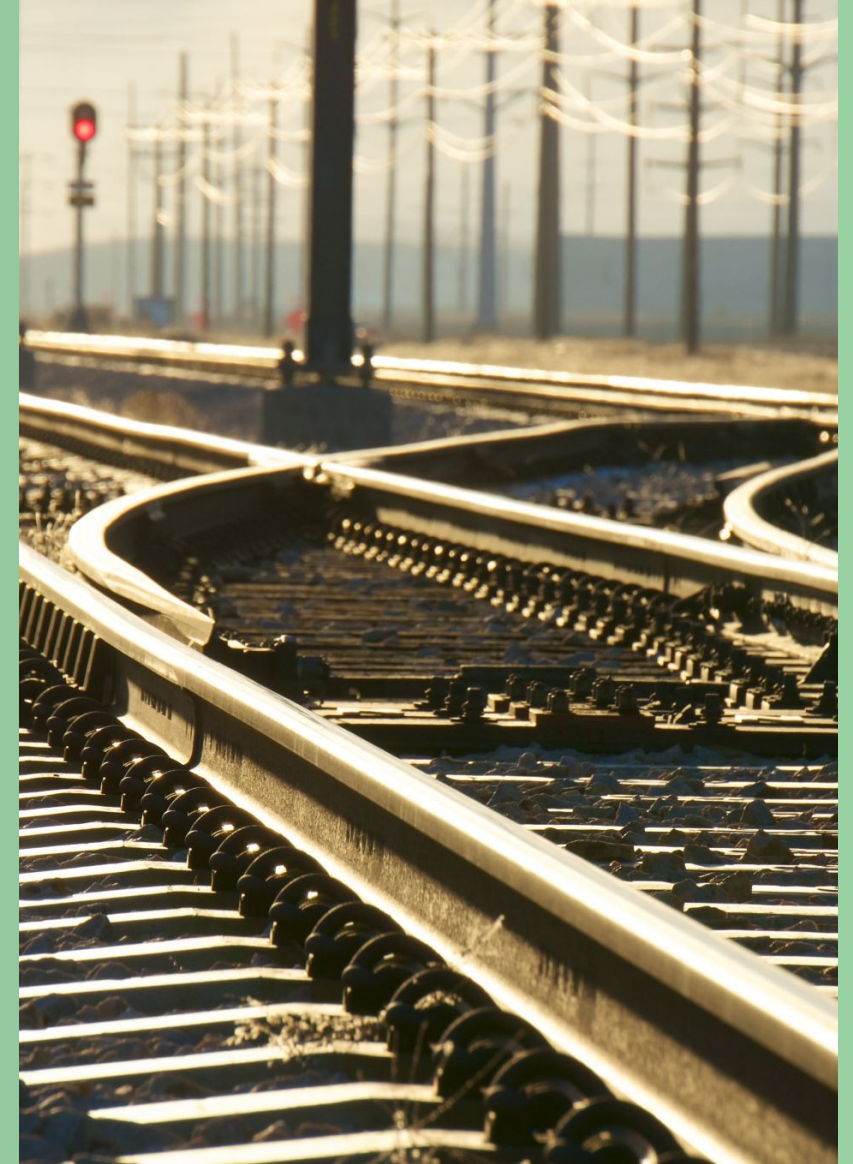
The Lay of the Land – An Introduction to Rail Transit

Bryan Sooter – Senior Director of Standards and Rail Engineering, APTA

B O S T O N

Outline

- Rail Transit Modes
- Characteristics of Rail Transit Modes
- Classification of Rail Transit Systems
- Eras of Rail Transit
- Global Growth of Rail Transit
- Railroad Track & Equipment Classifications
- State of Good Repair
- Additional Resources



Rail Transit Modes – Streetcar

- Operate routes on streets in mixed traffic.
- Typically operates with one- or two-car trains powered by overhead catenaries and has frequent stops.



Toronto Transit Commission Streetcar

Rail Transit Modes – Light Rail Transit

- Operates in mixed traffic or exclusive right-of-way (ROW)
- Train length of 1-4 cars
- Low or high platform
- Primarily overhead power



LA Metro Rail A-Line

Rail Transit Modes – Light Rail Rapid Transit

- Fully grade separated with light rail vehicles
- Used on lines with that require high performance but do not have high passenger volumes to justify large stations and long trains
- Examples include:
 - OC Transpo Line 1 (Confederation Line)
 - St. Louis Metro



SEPTA's Norristown High Speed Line



*The key thing with the **light metro** is that it's gonna be high speed and it's gonna have a lot of capacity, IBX Project Executive*

Rail Transit Modes – Heavy Rail

- Operates in exclusive ROW
- Long trains of six to eight cars or more
- Travel relatively short distances
- Most standardized mode



Rail Transit Modes – Commuter/Regional Rail

- Can be electric, diesel, or alternative fuel propelled
- Operates between a central city & outlying areas at higher average speeds than other rail modes
- Operates on the general railroad network (on both exclusive passenger corridors and on freight owned corridors)



METROLINK



Metra



Rail Transit Modes – Intercity & High-Speed Rail

Maximum speed on most intercity lines is 79 mph (130 kph)

Higher-Speed Rail (HrSR)

80-150 mph (130-250 kph)

High Speed Rail (HSR)

Greater than 150 mph (250 kph)



NextGen Acela



Transit Modes Characteristics

Transit modes are defined by three basic characteristics:

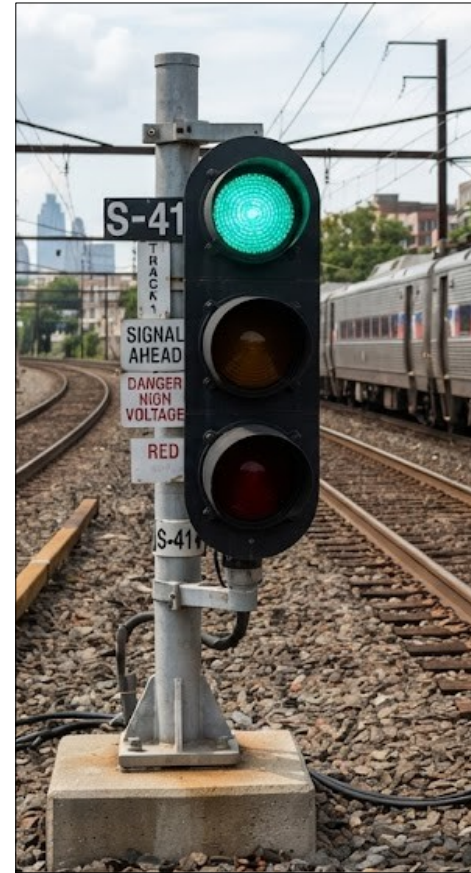
- **Right-of-Way (ROW) Category**
 - Category C – Surface Streets with Mixed Traffic
 - Category B – Longitudinally physically separated by curbs and barriers from other traffic
 - Category A – fully controlled ROW
- **System Technology**
 - Propulsion and control
- **Type of Service**
 - Local, peak, express



Train Control Systems

General Categories for Train Movement Control

- **Manual / Visual** – accepted as sufficient for Streetcars
- **Manual / Signals** – manual driving is assisted by automated signals showing occupancy of the track ahead also known as Automatic Train Protection (ATP)
 - Fixed block and track circuits are the historical method for ATP
 - Can be permissive or “forced stop”
- **Automatic Train Operation (ATO)** – driver initiates an automatic process and only supervises movement
- **Fully automated** – No personnel require on board for operation



Traction Power

Diesel-Electric— Used by primarily by commuter and intercity passenger railroads with ac traction motors

Pros: Range, versatile

Cons: slow acceleration, emission requirements

Alternative Fuel Source (hydrogen fuel cell or battery electric) -

Pros: Zero Emissions

Cons: New technology, limited applications, fuel/power availability



Metro North SC42-DM



Stadler FLIRT ZEMU

Traction Power

Overhead Catenary – Most proven and widely used system for electrified intercity rail, light rail, and high-speed railways. Voltages range up to 25 kV ac

Pros: reduced cost of power supply equipment and improving efficiency, increased safety at track level

Cons: Need for regular catenary maintenance and challenges from urban environments. Increased cost to tunnel construction

Third Rail - Primarily used on heavy rail systems at 750 V dc

Pros: No electromagnetic interference issues, reduced maintenance

Cons: Speed limited to ~100 mph, track maintenance, power gaps

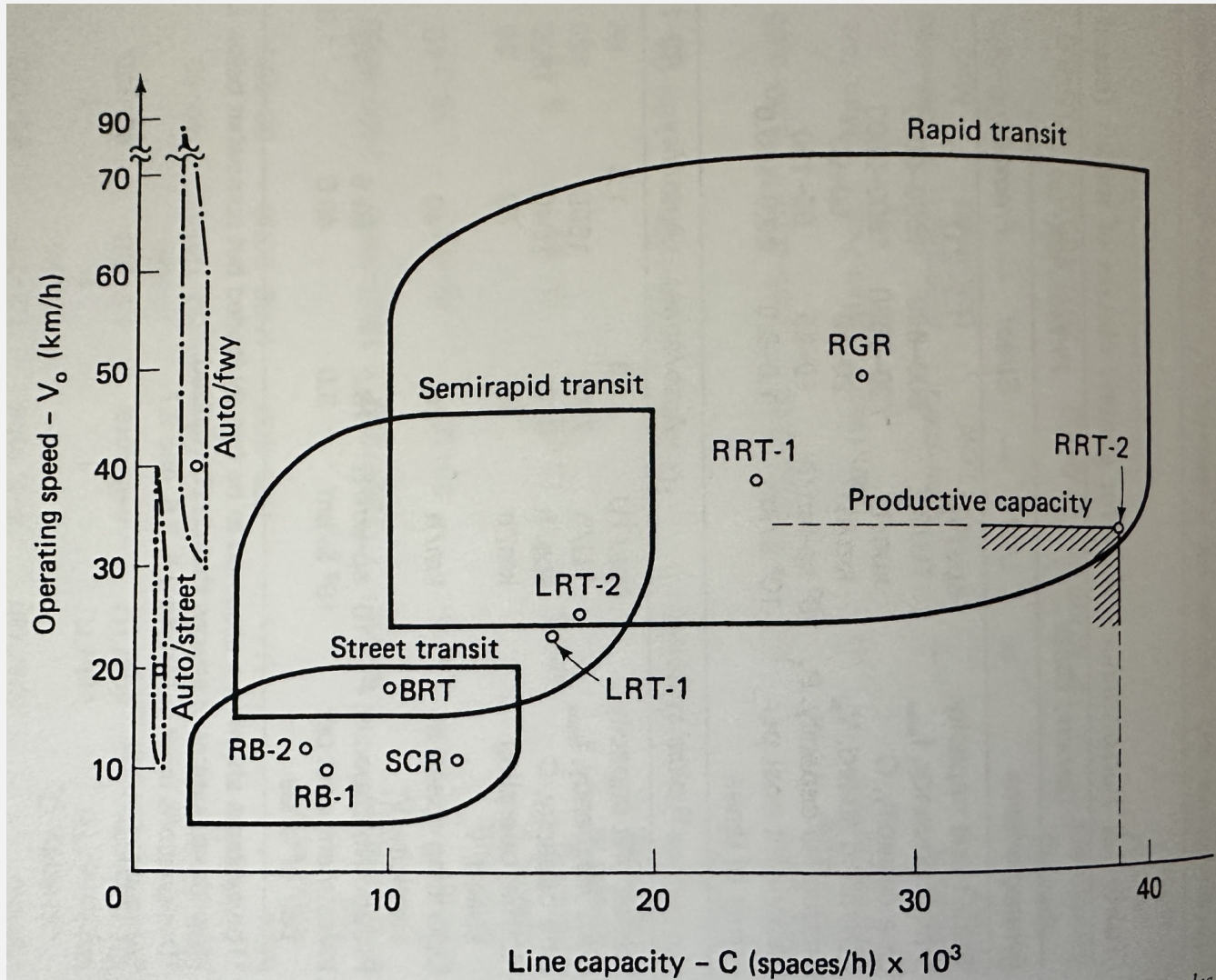


Classification of Urban Rail Transit Systems

Characteristics	Type \ Mode	Tramway - Streetcar	Light Rail	LRRT	Rapid Transit	Regional Rail
Right-of-way separation	None Some Grade crossing only Full	— — — —	— — — —	— — — —	— — — —	— — — —
Max no. of cars/train	1-3 4-10	— —	— —	— —	— —	— —
Station platform	Low High	— —	— —	— —	— —	— —
Power pick-up	Overhead Third rail (Diesel)	— — —	— — —	— — —	— — —	— — —
Vehicle travel control	Driver/visual Permissive signals Forced stop signals Automatic	— — — —	— — — —	— — — —	— — — —	— — — —
Max. vehicle speed	≤ 70 km/h 71-100 km/h > 100 km/h	— — —	— — —	— — —	— — —	— — —

From Vuchic 2007

Classification of Urban Rail Transit Systems



Line Capacity, C is the maximum number of people that can be carried past a point in one hour

From Vuchic 2007

Rail Transit Era's – Legacy Systems (pre WWII)

- 1892 Chicago “L”
- 1897 Boston (tunnels used by streetcars)
- 1904 New York City (trains from Manhattan to the Bronx)
- 1907 Philadelphia Market Street Line
- 1908 Hudson & Manhattan Railroad (PATH)
- 1910 New York Penn Station Opens



Main Concourse New York Penn Station – 1962

Rail Transit Era's – Rebirth of Rail Mass Transit

- 1954 – Toronto TTC Yongue Line
- 1957 – Cleveland RTA Red Line
- 1972 – Bay Area Rapid Transit
- 1976 – Washington DC Red Line
- 1978 – Edmonton (1st modern light rail system in North America)
- 1979 – Atlanta MARTA
- 1983 – Baltimore Subway Link
- 1984 – Miami Metrorail
- 1986 – Vancouver Skytrain
- 1986 – Portland Metropolitan Area Express (MAX)



Dupont Circle Station Construction

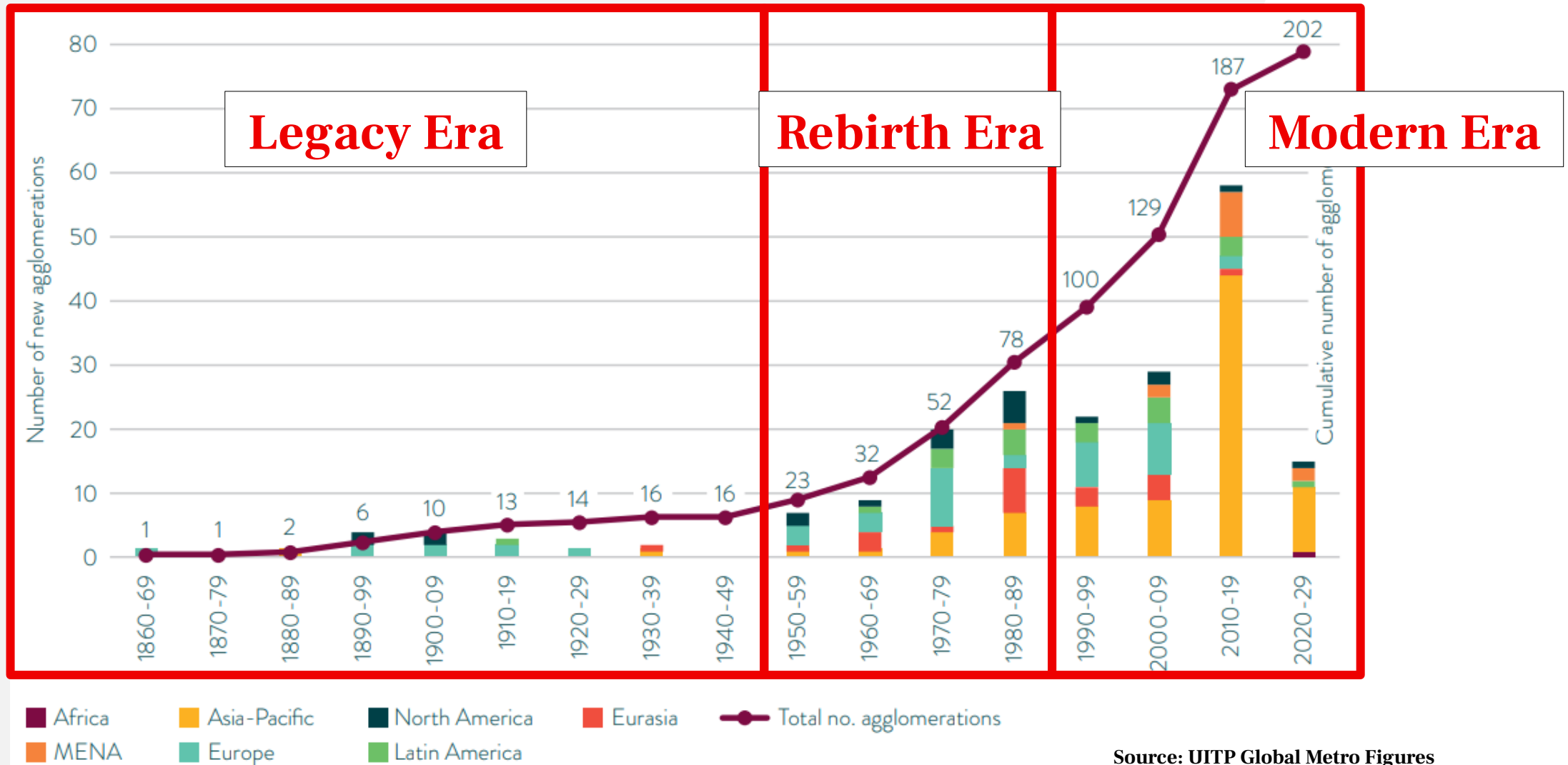
Rail Transit Era's – Modern Era

- 1990 – LA Metro Rail
- 1994 – Denver RTD Light Rail
- 1996 – Dallas DART Red & Blue Lines
- 1999 – Utah Transit Light Rail
- 2000 – New Jersey Transit Hudson-Bergen Light Rail
- 2000 – Acela Express (up to 150 mph)
- 2004 – Metro Transit Light Rail (Minneapolis)
- 2009 – Seattle Central Link
- 2019 – TexRail (DFW to Ft. Worth)
- 2023 – Réseau express métropolitain (REM)



Réseau express métropolitain (REM)

Global Growth of Metro Systems Over Time



Source: UITP Global Metro Figures
2024 (May 2025)

Railroad Track Classifications

	The maximum allowable operating speed for freight trains is—	The maximum allowable operating speed for passenger trains is—
Excepted track	10	N/A
Class 1 track	10	15
Class 2 track	25	30
Class 3 track	40	60
Class 4 track	60	80
Class 5 track	80	90
Over track that meets all of the requirements prescribed in this subpart for—	The maximum allowable operating speed for trains is 1	
Class 6 track	110 m.p.h.	
Class 7 track	125 m.p.h.	
Class 8 track	160 m.p.h.2	
Class 9 track	220 m.p.h.2	

**Grade crossing at Class 7 require FRA approval.
No at-grade (level) crossings on Class 8 and 9 track**

Passenger Rail Equipment Classifications



Tier I (conventional):

Maximum Speed = 125 mph (~200 kph)

Traditional “FRA Compliant” Equipment (49 CFR §229 & 238)



Tier II (Northeast Corridor):

Maximum speed = 160 mph (~260 kph)



Tier III (High Speed Rail)

Maximum Speed = 220 mph (355 kph)

Interoperable with all tiers up to 125 mph

Note: The FRA regulations that make up Tier III requirements was broken up into two rules informally known as NPRM 1 and NPRM 2.

Final Rule for NPRM 1 Published on November 21, 2018.

- Trainset
- Structure/Crashworthiness
- Interior Attachments
- Glazing
- Brake System

No final rule for NPRM 2. (The NPRM was published on April 3, 2023)

- Inspection Testing & Maintenance
- Compliance Testing & Startup
- General Safety
- Safety Appliances

State of Good Repair: The Operating Standard

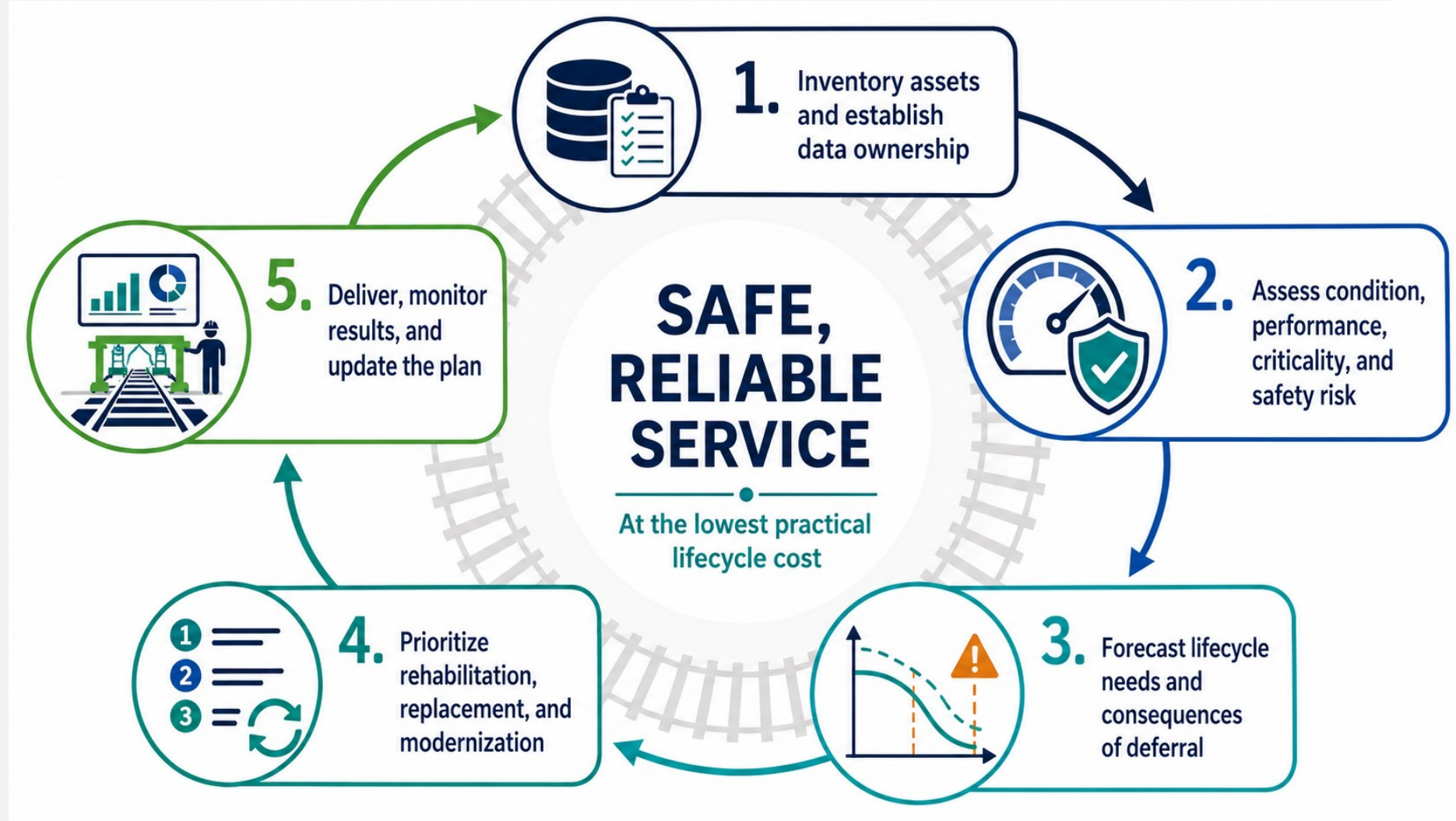
- An asset can perform its designed function
- Its use does not pose a known unacceptable safety risk
- Its lifecycle investment needs have been met or recovered
- SGR is a service outcome—not simply an asset's age



SGR Covers the Entire Rail Asset Portfolio



SGR Investment Is a Continuous Cycle



Additional Resources

Federally Sponsored Research Reports

Transport Canada

- <https://tc.canada.ca/en/rail-transportation/publications>

Federal Transit Administration

- <https://www.transit.dot.gov/research-innovation/fta-reports-and-publications>

Federal Railroad Administration

- <https://railroads.dot.gov/research-development/program-areas/human-factors/ctil/reports-presentations>



<https://www.apta.com/research-technical-resources/standards/>



<https://www.arena.org/>



<https://www.trb.org/TCRP/TCRP.aspx>